

Placement Experience

Preparation steps I followed:

- Coding - arrays, string(both are almost sufficient for online assessment), commonly asked graph(bfs, dfs based), trees(traversals), DP(easy,medium), DSA - linked list, stack, queue(given operations and find resulting output in OA).
- Used ACM CEG resources that were shared for summer internship and placement. Try solving all easy and medium problems. For arrays and strings you can try hard also.
- Coding websites - leetcode, gfg, hackerrank. Participate in leetcode/other online contests and ACM weekly contests for time management. ACM materials and contests are very useful.
- Resume - clear overview of everything mentioned(don't put false information).
- Projects - algo used, why u used it, what's the scope and use of the project, what's your contribution, challenges faced if any and how you overcame. I understood that a good project and tech stack we have worked on might also matter.
- Tell the interviewer if you don't know any concept and you'll learn about it. Or just say whatever you know(even basic things).
- Have a gist on Skills mentioned in the resume - basic/commonly asked interview questions. Some interviewers ask preferred coding language - then ask lang specific questions.
- Sorting algorithms - simulation, time complexity, space complexity (For OA MCQ).
- FYP - what, modules used etc. (definite ques on Fyp mostly during the interview).
- Go through common HR questions and have a good idea about the company, their values etc. before attending their interview.
- Refer to gfg articles before an interview for any company. But don't expect similar questions as everything depends.

Companies I attended for interview and questions I remember:

- Software AG - 2 tech rounds - system design - design a theatre booking website, question on image processing(given an image how to separate components in it based on colour).
- CitiBank - 1 round - Tech + HR - About Fyp, Projects, HR questions(I applied for EIOT so mostly simple questions were asked).
- Wells Fargo - 2 rounds -
For round 1, Queue based implementation ques(insert, delete from middle), Fyp and other old projects - scenarios were given and what will happen in the project was asked.

- Natwest - 1 round - Projects, DBMS - cursor, joins, sql queries, HR questions.
- Barclays - 1 round - 2 simple logic for string, list based question (something with conversion of case), tower of hanoi logic, Projects, Why I chose Engineering, having taken Biology in 12th?, My Inspiration etc.for HR(Cover RISES from their values).
- Bank of America - 2 rounds -
For round 1, Merge sort in code pair, Python - my preferred lang - so common ques like data types and about a ML project, garbage collection in java, DBMS concepts - sql queries, indexing.
- CPCL - 1 round - My advice is don't attend - They asked questions from a notebook and expected the same answers as it was there - About version control systems, ios vs Android diff, SQL commands, C vs C++ etc.
- Technip - 1 round - OA was easy but Interview takes into the role GET(They said the work covers all departments and will be trained accordingly, so I did not attend the interview).
- IDFC - 2 Tech + HR rounds - Database concepts - relational vs Nosql, api, idea of AWS, String operations in python for given use case (As I told I use python), HR ques-volunteering/interests, Projects - based on fyp.

Online Assessment questions I remember:

Coding questions were almost all based on arrays, string only. Other DSA and aptitude were covered in MCQs. Few companies were language specific. The overview of questions asked in few companies that I have taken note of are,

- Oracle - MCQ+coding - Good subsequences, Rest API max transfer(standard question), LCS.
- Appian - To find the number of values in an array exceeding the threshold. To arrange the given data/ file structure in tree format - not sure - allowed only using Java.
- BNY - 4 questions - 1 easy+ 2 medium + 1 hard - Count analogous array, glide typing, maximum request count given the bandwidth, Given array of binary numbers take their complement and find max sum.
- Software AG - Largest possible number formed from an array, longest cycle length in undirected graph, sorting an array based on 2 attributes.
- Arcadia - word pattern matching count, max prod subarray satisfying perfect square rule.
- Citi Bank - aptitude + 2 easy array based questions.
- Vegrow - 3 coding questions(not sure exactly) - 1 question on Coin change DP.
- Wells Fargo - kth smallest element in an array, konigsberg bridge problem, sorting an array based on 2 attributes.
- Caterpillar - 1st round - English comprehension and aptitude.

- Natwest - Max energy product's sum(sort the array and find the prod of 2 elements on each extreme, return the max), uncommon elements in two arrays.
- Dover - 60 mcq - Aptitude, coding and logic based questions
- ARM - Coding to be solved in C/C++.
- Accolite - MCQ, 2 coding - based on graphs.
- Verizon - 2 coding, 2 questions to give an english comprehension for a given use case.
- Societe Generale - 2 questions - OOP based coding round.
- Accenture - 1st round - MCQ, verbal, excel, aptitude - mock tests will be available online if needed, 2nd round - 2 medium coding questions based on array and strings.
- Barclays - MCQ(DSA, OOPS, OS, aptitude, output based) + 1 sql + 1 coding - lexicographically smallest string.
- IDFC - MCQ(tech+aptitude), 2 coding questions - 1 based on binary numbers - don't remember exactly.

References I used:

- Resources shared by ACM-CEG - [ACM_Materials](#)
- Youtube - striver for coding, neetcode, abdul bari - for time complexity related concepts, jeni's lecture - for DSA(for understanding how they work), neso academy - for theoretical concepts.
- Indiabix - for aptitude - time and distance, ratios, profit loss, ages, probability, time and work, averages, ap and gp, simple and compound interests are mostly covered.
- Top 25 questions of common topics(interviewbit, gfg) - C++, Java, DMBS - sql, normalisation, joins, OS, OOPS - 4 pillars.

About the company and role: IDFC FIRST Bank - Technology led banking, know the key values, vision, mission, some products etc. The role is Application Engineer - software development - major work covered in Golang, Java, Python, ReactJS, React Native. DevOps roles can also be assigned.

My Experience: Prepare consistently and don't lose hope. It is really hurting seeing the rejections even after much preparation, but the efforts you put in will pay off one day. Sometimes we miss tests because of time clashes and sometimes we can't get shortlisted for interviews even if we perform well in online tests/not get selected even if we think we performed well in the interview . But concentrate on the next one and try not to miss any opportunity from your side. Note down the mistakes you figure out as you attend interviews and try not to repeat them. Attend mock interviews conducted by our college. Try getting a summer internship as it is a great opportunity(Not much expectations from the company side as that of placement). Try your best and All the best!